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Guidelines for Treatment of *Helicobacter pylori* in Adults

Initial Testing: *Helicobacter pylori* testing should be performed in patients with active peptic ulcer disease (PUD), a history of PUD, dyspepsia, low-grade gastric mucosa-associated lymphoid tissue (MALT) lymphoma, a history of early gastric cancer endoscopic resection, chronic non-steroidal anti-inflammatory drug (NSAID) or aspirin use, unexplained iron deficiency anemia, or idiopathic thrombocytopenic purpura.

Who to Treat: All patients who test positive for an active *H. pylori* infection should be offered treatment. Success of *H. pylori* eradication depends on choice of regimen, patient adherence, and susceptibility to antibiotics. Antibiotic susceptibility is significantly dependent on previous exposures, and clarithromycin- or fluoroquinolone- based regimens should be avoided in patients previously exposed to these agents.

Test for Eradication: Test for eradication after *H. pylori* treatment should be performed with a urea breath test, fecal antigen test, or biopsy-based test at least 4 weeks after completion of treatment and after proton pump inhibitor (PPI) is held for 2 weeks.

Susceptibility Testing: No routine susceptibility testing is necessary prior to use of bismuth quadruple therapy, rifabutin triple therapy, potassium-competitive acid blocker dual therapy, or high-dose dual therapy.

In patients with infection refractory to the above and in whom a gastric biopsy is being performed, the MiChart order 'Helicobacter Pylori Culture and Susceptibility' should be ordered (https://www.pathology.med.umich.edu/handbook/#/details/877). This will provide results for amoxicillin, clarithromycin, levofloxacin, metronidazole, rifampin, and tetracycline. It typically takes 3-7 days to culture H pylori, and an additional 7-28 days for susceptibilities to result.

If H pylori does not grow on culture, a stool sample may be sent for clarithromycin genotypic testing (https://www.mayocliniclabs.com/test-catalog/overview/607594). This should be ordered as a 'miscellaneous send out': Helicobacter pylori with clarithromycin resistance prediction, PCR, feces (Mayo HPFRP). This will inform whether clarithro-containing concomitant therapy can be attempted. Another option in multiply refractory patients with significant prior treatment history is genotypic testing to clarithromycin, fluoroquinolones, metronidazole, amoxicillin, rifabutin, and tetracycline (https://amlaboratories.com/service/h-pylori-testing-services). This test is more expensive and as such should be reserved for scenarios where the other options are not sufficient and a regimen tailored to susceptibility is necessary. Note that consistent correlations between resistance and failure have been identified for quinolones and macrolides but not for other agents. This should be ordered as a "miscellaneous send out" order with notification to the Microbiology Attending on call (search for 'Microbiology Attending' on paging website).

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Initial Therapy	Salvage Therapy	Comments
Clarithromycin- and Levofloxacin- based regimens are not preferred treatments due to high prevalence of resistance. Resistance to these agents is associated with treatment failure. Regimens containing these agents is only recommended when susceptibility testing is performed and confirms susceptibility. Treatment duration of 14 days is preferred due to improved eradication compared to shorter treatment durations. <u>Preferred:</u> **Bismuth quadruple therapy** [1] Alternative: **Rifabutin triple therapy[1]** **OR** **Potassium-competitive acid blocker dual therapy[1]** <u>Preferred treatment in patients with a **verified** penicillin allergy:</u> *Patients with a reported penicillin allergy should be evaluated by the Beta-Lactam Allergy Evaluation Service (BLAES) (inpatients) or Allergy clinic (outpatients). > 90% of patients with a purported penicillin allergy are NOT allergic.* **Bismuth quadruple therapy[1]** <u>Treatment in patients who are NPO:</u> *Treatment of H pylori is never urgent, and thus intravenous therapy should almost never be considered. Consult Gastroenterology or Antimicrobial Stewardship for questions in special circumstances.*	Select a treatment regimen with different antibiotics than used in initial therapy. Treatment duration of 14 days is preferred. <u>If a clarithromycin-based regimen was used for initial therapy:</u> **Bismuth quadruple therapy**[1] OR **Rifabutin triple therapy[1]** <u>If bismuth quadruple therapy was used for initial therapy:</u> **Rifabutin triple therapy[1]** OR **Potassium-competitive acid blocker dual therapy[1]** <u>Other options (levofloxacin and clarithromycin- inclusive regimens should only be used in patients with confirmed susceptibility):</u> **Levofloxacin triple therapy** - Levofloxacin 500 mg QD* - Amoxicillin 1 g BID* - PPI standard dose BID **Concomitant therapy** - Clarithromycin[4] 500 mg BID* - Amoxicillin 1 g BID* - Metronidazole 500 mg BID or TID - PPI standard dose[3] BID **High-dose dual therapy** - Amoxicillin 1 g TID or 750 mg QID* - Potassium-competitive acid blocker BID or PPI standard or double dose TID or QID	- Treatment of H pylori is never urgent, and as such inpatient initiation of treatment is often unnecessary - PPI standard dose: o dexlansoprazole 30 mg o esomeprazole 20 mg o lansoprazole 30 mg o omeprazole 20 mg o pantoprazole 40 mg o rabeprazole 20 mg. - **Clarithromycin and rifabutin have significant drug-drug interaction potential, and interaction screens (e.g., using the Lexi-Comp interaction tool) must be performed before prescribing a regimen containing these agents**. Clarithromycin is a strong CYP3A4, OATP1B1/1B3, and P-glycoprotein/ABCB1 inhibitor. Rifabutin is a weak CYP2C9 and moderate CYP3A4 inducer. Significant drug-drug interactions may exist. - **Doxycycline** 100 mg BID substitution for **tetracycline** 500 mg QID in bismuth quadruple therapy has not been robustly studied in the U.S. Limited studies suggest doxycycline may be less effective than tetracycline. - For patients on antibiotics for a different infection, the following antibiotics are effective *in vitro* but have not been robustly studied *in vivo*: penicillins (ampicillin, piperacillin), cephalosporins (cefixime, cefuroxime, cefotaxime, ceftazidime), carbapenems (imipenem, meropenem). Treatment with a preferred regimen containing 2-3 antibiotics is recommended to increase eradication success. * Dose may need to be adjusted for renal dysfunction. See Michigan Medicine DOSING GUIDELINES

FOOTNOTE 1:

- **Bismuth quadruple therapy** is available as:
    a. Pylera branded product. 3 capsules (bismuth subcitrate 420 mg, metronidazole 375 mg, tetracycline 375 mg) QID with PPI standard dose BID (~$80/day). A generic formulation is also available (~$55/day).
    b. Component regimen: Bismuth subsalicylate (most commonly using Pepto-Bismol) 262mg QID, Metronidazole 500 mg TID or QID, Tetracycline 500 mg QID, PPI standard dose BID
    c. Component regimen is an appropriate substitution for combination products and is preferred for inpatient use. However, treatment can almost always be deferred until the patient is discharged.
- **Rifabutin triple therapy** is available as:
    a. Talicia branded product 4 capsules (omeprazole 40 mg, amoxicillin 1000 mg, rifabutin 50 mg) every 8 hours (~$50/day)
    b. Rifabutin 300 mg QD, Amoxicillin 1 g BID*, PPI standard dose BID
    c. Given greater rifabutin intragastric exposure, and higher amoxicillin and PPI doses with Talicia, this product is preferred (see Howden reference) to the component regimen.
- **Potassium-competitive acid blocker dual therapy** is NOT on inpatient formulary. It is available for outpatients as:
    a. Voquezna DualPak branded product: Vonoprazan 20 mg BID and Amoxicillin 1,000 mg TID
- For patients planned to be discharged on a combination product, outpatient coverage should be verified by ordering 'Consult to Transitions of Care Pharmacy (TOC) Adult'

Last revised: 11/2024	**Antimicrobial Stewardship Subcommittee:** 10/2024	**P&T:** 11/2024

References

	Citation
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Revision History

Date	Revision
11/2024	Added detail on susceptibility testing, revised treatment recommendations

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source